

The formula (given in formula book):

$$\int u \frac{dv}{dx} dx = uv - \int \frac{du}{dx} v dx$$

One part is differentiated

The other part is integrated.

C4 Integration by Parts

Integration by parts is used to integrate a product. One part of the product (u) should be easy to differentiate (and will usually simplify when differentiated). The other part $\left(\frac{dv}{dx}\right)$ should be easy to integrate and should not become too much harder when integrated.

Example 1: Find $\int 2xe^x dx$.

This is a suitable candidate for integration by parts

with $u = 2x$ and $\frac{dv}{dx} = e^x$:

$$u = 2x \Rightarrow \frac{du}{dx} = 2$$

$$\frac{dv}{dx} = e^x \Rightarrow v = e^x$$

Substitute these into the formula:

$$\int 2xe^x dx = 2xe^x - \int 2e^x dx = 2xe^x - 2e^x + c$$

Example 2: Find $\int x \cos x dx$.

Here we take $u = x$ and $\frac{dv}{dx} = \cos x$:

$$u = x \Rightarrow \frac{du}{dx} = 1$$

$$\frac{dv}{dx} = \cos x \Rightarrow v = \sin x$$

Substitute these into the formula:

$$\begin{aligned} \int x \cos x dx &= x \sin x - \int \sin x dx = x \sin x - (-\cos x) + c \\ &= x \sin x + \cos x + c \end{aligned}$$

Note: Sometimes it is necessary to use the integration by parts formula twice (e.g. with $\int x^2 \sin x dx$).

Common examination questions

Example 1: Find $\int x \ln x dx$.

This can be found using integration by parts if we

take $u = \ln x$ and $\frac{dv}{dx} = x$.

$$u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$\frac{dv}{dx} = x \Rightarrow v = \frac{x^2}{2}$$

Substitute these into the formula:

$$\begin{aligned} \int x \ln x dx &= \frac{x^2}{2} \ln x - \int \frac{1}{x} \times \frac{x^2}{2} dx = \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x dx \\ &= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + c \end{aligned}$$

Example 2: Find $\int \ln x dx$.

This can be thought of as $\int 1 \ln x dx$ and so can be

integrated by parts with $u = \ln x$ and $\frac{dv}{dx} = 1$

$$u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$\frac{dv}{dx} = 1 \Rightarrow v = x$$

$$\begin{aligned} \int \ln x dx &= x \ln x - \int \frac{1}{x} \times x dx = x \ln x - \int 1 dx \\ &= x \ln x - x + c \end{aligned}$$

Definite integrals (using by parts)

Example: a) Find the points where the graph of $y = (2-x)e^{-x}$ cuts the x and y axes.

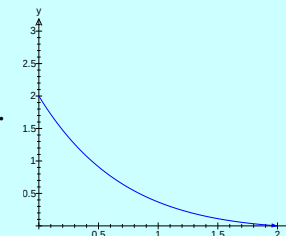
b) Sketch the graph of $y = (2-x)e^{-x}$.

c) Find the area of the region between the axes and the graph of $y = (2-x)e^{-x}$.

a) Graph cuts y -axis when $x = 0$, i.e. at $y = 2$

Graph cuts the x -axis when $y = 0$, i.e. when $x = 2$.

b) The graph looks like:



c) Area is $\int_0^2 (2-x)e^{-x} dx$.

$$u = 2-x \Rightarrow \frac{du}{dx} = -1$$

$$\frac{dv}{dx} = e^{-x} \Rightarrow v = -e^{-x}$$

$$\begin{aligned} \text{So } \int (2-x)e^{-x} dx &= (2-x)(-e^{-x}) - \int -1 \times -e^{-x} dx \\ &= -(2-x)(e^{-x}) - \int e^{-x} dx \\ &= -(2-x)(e^{-x}) + e^{-x} \end{aligned}$$

Now that we've integrated, we substitute in our limits:

$$\begin{aligned} \int_0^2 (2-x)e^{-x} dx &= \left[-(2-x)e^{-x} + e^{-x} \right]_0^2 \\ &= (-0e^{-2} + e^{-2}) - (-2e^0 + e^0) \\ &= 0.135 + 1 = 1.135 \end{aligned}$$

